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REVIEW



Behavioural Therapy for tic disorders: a comprehensive review of the literature

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ABSTRACT

Introduction: Behavioral therapies are recommended as a first-line intervention for Tourette syndrome and persistent motor or phonic tic disorder.

Areas covered: In this review, the authors summarize randomized controlled trials on the comprehensive behavioral intervention for tics (CBIT), habit reversal therapy (HRT), and exposure and response prevention (ERP). Studies of face-to-face treatment, treatment by video conferencing, group treatment, and internet delivered treatment were assessed, as well as evidence of treatment predictors, modifiers, and mediators.

Expert opinion: There is high-quality evidence for face-to-face one-on-one treatment with CBIT, and data suggesting that one-on-one treatment by videoconference provides similar benefit. Limited data on group treatment with CBIT/HRT suggests inferiority to individual treatment, while internet-based CBIT programs appear more beneficial than wait list or psychoeducation. There is one face-to-face one-on-one treatment comparison of ERP to HRT, suggesting equal benefit. Internet-based ERP with minimal therapist support appears effective, although effect sizes are small. One study using behavioral therapy with ERP or HRT found similar benefit to medical treatment with antipsychotics. Data on predictors, modifiers, and mediators of treatment efficacy are emerging. In summary, behavioral therapies are an important treatment modality for tic disorders. Furthermore, important efforts to improve treatment accessibility are underway.

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Tourette syndrome; tic disorder; behavioral therapy; comprehensive behavioral intervention for tics; exposure and response prevention; habit reversal therapy

1. Introduction

The use of behavioral therapies for tic disorders was first reported in the 1970s by Azrin and Nunn, who described using habit reversal for eliminating nervous habits and tics [1]. Over the past five decades, research on the efficacy of behavioral therapies for tics, primarily the comprehensive behavioral intervention for tics (CBIT) and exposure and response prevention (ERP), has steadily increased. This led to practice guideline recommendations from the American Academy of Neurology in 2019 and the European Society for the Study of Tourette Syndrome in 2022 that CBIT be offered as a first-line intervention to children and adults with tics, given high confidence in the evidence for efficacy [2,3]. In recent years, research on behavioral therapy for tics has focused on administering both CBIT and ERP remotely by videoconference, through the use of internet-based platforms with minimal therapist support, or in a group setting.

Despite their effectiveness, the underlying mechanisms of behavioral therapies remain unclear. Several processes have been suggested as potential mechanisms of behavioral therapy [4,5]. For instance, cognitive control, which allows the organization of cognitions and actions according to task demands as well as the suppression of inappropriate actions [6], is thought to be involved in behavioral therapy for tics [7]. Learning processes, such as associative learning, could also play a role. For instance, CBIT may

work by breaking the negative reinforcement cycle between premonitory urges and tics and by creating new associations between premonitory urges and competing responses [8]. For ERP, one premise is that exposure to premonitory urges without performing tics would eventually lead to habituation to the urge [7]. Yet, evidence supporting such processes as mechanisms of behavioral therapy is scarce. In this review article, we will describe the key evidence from randomized controlled trials (RCT) which established the efficacy of CBIT and ERP for the treatment of tic disorders, including a description of study risk of bias and demonstrated effect sizes. Study risk of bias is described using the American Academy of Neurology system [9] (see Table 1), which rates studies between Class I and IV based on trial characteristics. We will then discuss research on predictors, moderators, and mediators of efficacy of behavioral therapy for tics, which may help us understand how behavioral therapy works and for whom it might work best. Finally, we will discuss challenges with access to behavioral therapy, efforts to establish the efficacy of remote delivery systems to improve the availability of this treatment modality, and available books or online programs which endeavor to teach patients how to use this therapy. We believe that this review brings together information on the efficacy, mechanisms, and availability of behavioral therapies for tics that is essential for clinicians to make informed decisions on recommending its use to people with tics seeking treatment.

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Article highlights

- Behavioral therapy for tics has received extensive empirical support and is considered as a first-line treatment for Tourette syndrome and other tic disorders.
- Behavioral therapy and pharmacotherapy were found to have similar impacts on tic severity.
- Several barriers, such as a lack of therapists, especially in remote areas, as well as the costs of treatment, may hinder access to behavioral therapy.
- New developments in group-based treatment, remote therapy, therapist-, and parent-guided behavioral interventions, and self-help programs could improve access to treatment.
- Several psychological processes have been suggested as mechanisms of behavioral therapy, but evidence supporting those mechanisms remains scarce.

2. Evidence for the use of behavioral therapies for tics

2.1. Comprehensive behavioral intervention for tics/habit reversal training

2.1.1. Face-to-face, individual treatment

CBIT is a multifaceted intervention. Habit reversal training is the foundation of CBIT. It involves developing tic awareness

through self monitoring of tics and premonitory urges, and competing response training, where one engages in a voluntary behavior that is incompatible with the tic when the urge arises. Other components of CBIT are relaxation training and a functional behavioral assessment to identify situations which influence tic severity, and develop strategies to reduce the influence of these factors.

Evidence for the efficacy of CBIT mainly stems from two large Class I RCTs in 126 youths and 122 adults with tics. In both studies, participants were randomized to either CBIT or supportive therapy and psychoeducation for eight sessions over a period of 10 weeks. In the youth study [10], total tic severity on the Yale Global Tic Severity Scale Total Tic Score (YGTSS TTS) decreased from 24.7 points at baseline to 17.1 points at week 10 with CBIT, compared to a decrease from 24.6 points to 21.1 points with supportive therapy (SMD, 0.51 [95% CI, 0.15–0.86]). Worsening of tics was reported in one participant receiving CBIT and four participants receiving supportive therapy. Comorbid conditions were frequent, and 36.5% were taking medication for tics. No serious adverse events occurred during the study. Importantly, at 6-month follow-up, 86.9% of the participants receiving CBIT remained treatment responders. In a long-term follow-up study of 80 of

Table 1. Summary of American Academy of Neurology Criteria for risk of bias in therapeutic studies. Reproduced from [9] with permission of the American Academy of Neurology.

Class I Criteria
Randomized controlled clinical trial (RCT) in a representative population
Triple-masked studies (i.e. the patient, treating provider, and outcome assessors are unaware of treatment assignment)
Relevant baseline characteristics of treatment groups (or treatment-order groups for crossover trials) are presented and substantially equivalent between treatment groups, or there is appropriate statistical adjustment for differences
Additional Class I criteria:
a. Concealed allocation
b. No more than two primary outcomes specified
c. Exclusion and inclusion criteria clearly defined
d. Adequate accounting of dropouts (with at least 80% of participants completing the study) and crossovers
e. For noninferiority or equivalence trials claiming to prove efficacy for one or both drugs, the following are also required*:
i. The authors explicitly state the clinically meaningful difference to be excluded by defining the threshold for equivalence or noninferiority
ii. The standard treatment used in the study is substantially similar to that used in previous studies establishing efficacy of the standard treatment (e.g. for a drug, the mode of administration, dose, and dosage adjustments are similar to those previously shown to be effective)
iii. The inclusion and exclusion criteria for participant selection and the outcomes of participants on the standard treatment are comparable with those of previous studies establishing efficacy of the standard treatment
iv. The interpretation of the study results is based on a per-protocol analysis that accounts for dropouts or crossovers
v. For crossover trials, both period and carryover effects are examined and statistical adjustments performed, if appropriate
Class II Criteria
RCT that lacks one or two Class I criteria a – e (see above)
Cohort studies employing methods that successfully match treatment groups on relevant baseline characteristics (e.g. propensity score matching) meeting Class I criteria b – e (see above)
Randomized crossover trial missing one of the following two criteria:
a. Period and carryover effects described
b. Baseline characteristics of treatment order groups presented
All relevant baseline characteristics are presented and substantially equivalent across treatment groups (or treatment order groups for crossover trials), or there is appropriate statistical adjustment for differences
Masked or objective** outcome assessment
Class III Criteria
Controlled studies (including studies with external controls such as well-defined natural history controls)
Crossover trial missing both of the following two criteria:
a. Period and carryover effects
b. Presentation of baseline characteristics
A description of major confounding differences between treatment groups that could affect outcome**
Outcome assessment performed by someone who is not a member of the treatment team
Class IV Criteria
Studies not meeting Class I, II, or III criteria

*Numbers i–iii in Class Ie are required for Class II in equivalence trials. If any one of the three is missing, the class is automatically downgraded to Class III.

**Objective outcome measurement: an outcome measure that is unlikely to be affected by an observer's (patient, treating physician, investigator) expectation or bias (e.g. blood tests, administrative outcome data).

the original trial participants 11.2 years later, tic severity decreased significantly over time, with 40% of participants reporting partial remission. CBIT responders ($n=21$) in the original trial were more likely (67%) to achieve remission at follow-up (Total Tic Score = 12.52, SD = 10.75) compared to psychoeducation/supportive therapy responders ($n=6$, 0%) at follow-up (Total Tic Score = 20.67, SD = 6.92) on the YGTSS [11]. In the adult study [12], CBIT was superior to supportive therapy and psychoeducation, as measured on the YGTSS TTS, with an SMD of 0.62 (95% CI, 0.25–0.98). Worsening of tics during the study was reported in four participants receiving CBIT and four participants receiving supportive therapy.

Prior to these studies, others have provided initial support for habit reversal therapy. Deckersbach et al. [13] conducted a Class III randomized, unblinded study in 30 adults of habit reversal therapy, compared with supportive psychotherapy. Fourteen sessions of therapy occurred over a 5-month period. The YGTSS TTS decreased from 29.3 points at baseline to 18.3 points post-treatment with habit reversal therapy, in comparison to 27.7 points to 26.6 points with supportive psychotherapy (SMD, 1.41 [95% CI, 0.62–2.22]). Ten of 15 participants receiving habit reversal training were ‘much improved’ or ‘very much improved’ at the end of the study, compared with 2 of 15 participants in the supportive psychotherapy group ($p=0.008$). Wilhelm et al. [14] conducted a Class III randomized unblinded study of habit reversal therapy compared with supportive psychotherapy in 32 adults. Fourteen sessions of therapy were provided over a 5-month period. Habit reversal therapy was more effective in improving tics than supportive psychotherapy, with an SMD of 0.85 (95% CI, 0.09–1.61) on the YGTSS TTS. Thus, CBIT/HRT, when delivered face-to-face and one-on-one with a therapist, is well supported by empirical evidence.

2.1.2. Group-based treatment

CBIT and HRT have been adapted for group-based treatment. The efficacy of group treatment with HRT compared to group psychoeducation was evaluated in a Class II study which randomized 33 children to eight sessions of either invention, plus four parent group sessions [15]. All group sessions were structured and manualized and involved group discussion, didactic teaching, and small group activities. Children randomized to the HRT group chose up to three tics to treat and were given detailed instruction on competing responses. The psychoeducational group was trained in cognitive-behavioral strategies including identification of triggers to anger and problem solving approaches to build self-esteem and organizational skills. Participants receiving HRT were found to have a larger reduction in motor tics after treatment, while no difference was found regarding phonic tics. Participants from both groups also demonstrated long-term improvement in tic severity, quality of life and improvement in school attendance that persisted over 12 months [16].

A Class II study in Israel [17] compared the efficacy of group-based CBIT to a group-based Educational Intervention for Tics (group-EIT). In this study, both impact on tics as well as comorbid mental disorders were evaluated. Assessment was done at

baseline, post-intervention, and after 3 months. All participants received the intervention weekly over a period of 8 weeks. In addition, they also received three monthly booster sessions. Both intervention sessions included didactic teaching, activities in small groups or pairs, free discussion, and team games. Children randomized to the group receiving CBIT chose up to three of their most bothersome tics for treatment. They received instruction on tic awareness/urge monitoring and competing response training, the functional intervention and relaxation training. Children randomized to the educational intervention received psychoeducation, relaxation training techniques, and participated in group discussions related to coping, improving self-esteem, and organizational skills. Motor tic severity on the YGTSS fell from 14.3 (SD 5.1) to 9.7 (SD 6.8) in the CBIT group post-intervention, an effect that was maintained 3 months postintervention, 10.9 (SD 6.2). In the educational intervention group, motor tic severity fell from 14.0 (SD 3.5) to 8.0 (SD 5.1) postintervention, but increased to 14.1 (SD 4.3) 3 months postintervention. In contrast, vocal tic severity significantly increased in both groups from pre-intervention to post-intervention and significantly decreased in both groups 3-month postintervention, resulting in a nonsignificant change compared to baseline. More evidence is, therefore, needed to determine whether group-based CBIT/HRT can be recommended as an equivalent alternative to one-on-one treatment.

2.2. Exposure and response prevention

2.2.1. Face-to-face, individual treatment

ERP, especially when administered face-to-face, has received less empirical support than CBIT/HRT. There is one Class II study comparing ERP to HRT in 43 children and adults with TS [18]. ERP treatment consisted of 12 weekly 2-h sessions, while HRT consisted of 10 weekly 1-h sessions with a trained psychologist. Both treatment groups had significant improvement in tic severity over the course of the study on the YGTSS TTS, with no difference between treatments in efficacy (SMD, 0.25 [95% CI, −0.40, 0.90]). Adverse effects of therapy were not described.

2.2.2. Group-based treatment

Scandinavian researchers developed a new behavioral treatment manual for tics combining HRT and ERP and tested the effectiveness of this intervention administered either individually, or in a group setting [19]. Fifty-nine children and youth were randomized in a Class II RCT to either individual or group treatment. Both treatments incorporated nine sessions. The content of sessions was similar in individual and group treatment, and consisted of psychoeducation, HRT training, and ERP training. Group sessions consisted of four participants and one therapist, lasting 2 h. Individual treatment sessions lasted an hour. There was a significant tic reduction in both treatment groups, with a mean difference from baseline to endpoint of 9.48 (SD 7.84) on the YGTSS TTS with individual treatment, compared to a mean difference of 7.48 (SD 5.44) with group treatment. Reductions in tic severity did not differ across groups. However, for functional impairment, a significantly greater reduction was shown with individual treatment. In

addition, a positive effect on tic severity persisted over the period of 12 months after both interventions [20].

2.3. Studies comparing behavioral therapy with pharmacotherapy

Whether behavioral therapy allows similar improvement compared with pharmacotherapy is an important question when deciding on which treatment to administer. To answer this question, Rizzo et al. [21] compared behavioral interventions with pharmacotherapy and psychoeducation for the treatment of tics. This was a 10-week Class II RCT in which patients were randomized to psychoeducation, behavioral therapy (either HRT or ERP), or pharmacotherapy. Both HRT and ERP were administered following the therapist manual by Verdellen et al. [22]. ERP or HRT sessions were conducted over a period of 8 weeks. Pharmacotherapy consisted of risperidone which was given to drug-naïve children and adolescents (dose between 0.5 and 2 mg). In cases when patients were treated with risperidone in the past, they received aripiprazole (dose between 2.5 and 10 mg). Finally, those who had been treated in the past both with risperidone and aripiprazole were treated with pimozide at a dose ranging between 2 and 5 mg. Altogether, 110 patients were enrolled, with 26 receiving behavioral therapy, 57 receiving pharmacotherapy, and 29 receiving psychoeducation. Tic reduction was equal in the behavioral therapy and pharmacotherapy groups and significantly greater than in the psychoeducation group, with an 8.3 point reduction (95% CI 4.0–12.6) on the YGTSS TTS with behavioral therapy, and an 8.4 point reduction (95% CI 5.6–11.3) in the pharmacotherapy group. In comparison, the YGTSS TTS in the psychoeducation group decreased by 0.3 points (−3.6, 4.2). Whereas more evidence on this topic would be welcome, behavioral therapy appears to be as effective as pharmacotherapy in youth with TS.

3. How and for whom does behavioral therapy for tics work: treatment predictors, moderators, and mediators

The mechanisms through which behavioral therapy works remain poorly understood, and it is unclear if some individual characteristics may predict treatment outcome. The following section focuses on treatment predictors, moderators, and mediators. Predictors are variables predicting treatment outcome independently of treatment assignment, while moderators are variables predicting differential outcome according to treatment assignment [23]. Mediators are processes that are impacted by treatment, are associated with treatment outcome, and account for the association between treatment allocation and treatment outcome [23–25].

3.1. Symptoms and treatment characteristics

In RCTs of CBIT, several clinical characteristics have been identified as nonspecific predictors. For instance, greater tic severity and higher treatment expectations predicted larger reductions in tic severity for both CBIT and psychoeducation

and supportive therapy [26]. Additionally, more severe premonitory urges and co-occurring anxiety disorders predicted smaller decreases in tic severity independently of treatment allocation. In an RCT comparing individual vs. group combined CBIT/ERP treatment, the presence of obsessive-compulsive symptoms predicted a better treatment response irrespective of treatment allocation [27]. That study also reported that increased internalizing symptoms predicted reduced improvements in functional impairment.

In terms of moderators, CBIT was found to be effective regardless of medication intake or co-occurring disorders such as attention deficit hyperactivity disorder (ADHD), obsessive-compulsive disorder (OCD), or anxiety disorders [26]. Treatment outcomes appear to be moderated by the number of sessions, as a meta-analysis of behavioral therapy for tics reported slightly larger effect sizes in studies using more sessions [28]. Nissen, Parner [27] identified a few baseline characteristics indicating whether youths with TS benefited more from individual or group treatment. They found that youths with more severe internalizing and externalizing symptoms and those with sensory hypersensitivity fared better in the individual treatment, whereas those with more severe premonitory urges fared better in the group treatment. Thus, current evidence suggests that there is no clinical profile that is unsuitable for behavioral therapy, but some treatments may work better depending on individual profiles.

3.2. Cognitive control

Few studies have assessed whether cognitive control predicts or moderates behavioral therapy outcomes. Deckersbach, Rauch [29] found that inhibitory deficits in a visuospatial priming response inhibition task predicted smaller decreases in tic severity following HRT in adults. Another study, conducted in children, reported that better performance during the inhibition/switching condition of the Stroop task predicted larger reductions in tic severity after CBIT [30]. However, in both studies, associations between cognitive control and treatment outcome were not assessed across treatment conditions and could thus only be considered as nonspecific treatment predictors. Two studies stemming from the large RCTs of CBIT in children and adults reported that cognitive control processes did not moderate treatment outcome [31,32]. Similarly, in an RCT of CBIT vs treatment-as-usual, behavioral, and electrophysiological markers of cognitive control did not moderate CBIT outcome. Therefore, there are some preliminary evidence that cognitive control may constitute a nonspecific predictor of behavioral therapy outcome, but evidence of cognitive control acting as a treatment moderator is currently lacking.

Whether cognitive control mediates behavioral therapy outcome is not clear either. Petruo, Bodmer [33] reported improved response inhibition following CBIT in a sample of 21 adults with TS. However, those improvements in cognitive control were unrelated to the reduction in tic severity induced by CBIT and, therefore, cannot explain the change in symptoms. Another study found that putamen activation during a visuospatial priming response inhibition decreased from CBIT baseline to endpoint [34]. However, those decreases in

putamen activation did not predict treatment outcome; but decreases in inferior frontal gyrus activation did. Therefore, this study could not identify a specific mechanism mediating behavioral therapy. This study also included only eight adults with TS and its results have not been replicated yet. Furthermore, the latter two studies were not RCTs, so we cannot know to what extent those effects were specific to behavioral therapy. Two larger and more recent studies (both RCTs) found no impact of CBIT on any cognitive control measure [30,32]. CBIT also did not impact electrophysiological markers of cognitive control across the course of treatment [35]. Thus, currently available evidence suggests that cognitive control probably does not mediate behavioral therapy for tics.

3.3. Associative learning

Negative reinforcement, along with positive reinforcement, reversal learning, and extinction, are various forms of associative learning that have been hypothesized to be involved in behavioral therapy for tics [7]. However, whereas associative learning is hypothesized to underlie change in symptoms during behavioral therapy, evidence supporting this hypothesis is scarce.

To this date, only two studies have assessed associative-learning processes in the context of behavioral therapy for TS. For children and adults who took part in RCTs of CBIT, better homework (i.e. practicing CBIT skills at home between session) adherence predicted a better CBIT outcome [36]. This homework included a behavioral reward system, suggesting a potential impact of positive reinforcement on treatment outcome. Another study assessed whether d-cycloserine, which is known to enhance reversal learning and extinction, would optimize the effects of HRT [37]. It was found that children who received HRT + d-cycloserine had greater reductions in tic severity than those who only received HRT. While this study did not assess changes in associative learning processes, it reported that a medication with well-known effects on associative learning improved treatment outcome. This suggests that associative learning processes may constitute potential mediators of behavioral therapy outcome, but more research is needed.

3.4. Habituation

Even if habituation has been suggested as a mechanism of behavioral therapy for tics, only a few experimental studies tested this hypothesis in clinical studies. Initial studies of ERP suggested both within- and across-session habituation to premonitory urges [38,39]. However, a recent study with more fine-grained analyses revealed no within-session habituation during the ERP treatment [40]. In CBIT, decreased severity of premonitory urges across sessions was only found in adults who were considered as treatment responders [41]. This was not true in children, and across-session reductions in premonitory urges did not mediate treatment response for either children or adults. Another study collected ratings of premonitory urges across free-to-tic, tic suppression, and HRT conditions in adults [42]. In the HRT condition, participants were instructed to suppress a specific tic using a competing response. Most

participants did not report within-session reductions in premonitory urges during HRT. Thus, although some studies suggested possible changes in premonitory urges severity across and within treatment sessions, evidence to consider habituation as a mechanism of behavioral therapy for tics is currently lacking, especially given that no study found it to mediate treatment response.

3.5. Metacognitions

Metacognitions are an important feature in TS [43]. Nissen, Parner [27] found an association between reductions in thoughts about tics during behavioral therapy and a decrease in functional impairment after treatment. While this was not part of a mediation analysis, there may be hints that metacognitive aspects may underlie changes in perceived impairment during behavioral therapy for tics.

3.6. Motor functions

The role played by motor functions regarding response to behavioral therapy has mostly been assessed within O'Connor's cognitive-psychophysiological treatment model [44,45]. Studies have reported changes in movement-related neural activity [46,47] as well as improved motor skills following treatment [46,48]. Movement-related neural measures also predicted treatment outcome [47]. However, none of these studies were RCTs and they did not assess whether treatment-induced changes explained the improvement in tic symptoms. Therefore, we do not know if motor functions can be considered as moderator or mediator of treatment outcome.

3.7. Tic suppression

Behavioral therapies for tics have often been suggested to work by enhancing inherent tic suppression abilities (e.g. [49]). However, the involvement of tic suppressibility in behavioral therapy for tics has received little empirical support so far. For instance, even though ERP consists of guided tic suppression, studies assessing objective measures of tic suppressibility in a treatment context are scarce. Regarding CBIT, objective measures of tic suppressibility were not found to predict treatment outcome [30]. However, baseline functional brain connectivity during tic suppression was found to moderate CBIT outcome, so that children with increased functional connectivity at baseline showed larger decreases in phonic (but not motor) tic severity [50].

3.8. Treatment predictors, moderators, and mediators: future directions

The literature on behavioral therapy for TS is not extensive, especially when compared with other conditions such as depression or anxiety. Therefore, the search for processes predicting, moderating, or mediating treatment outcome is still in its early stage. We suggest that future treatment studies, and especially RCTs, systematically collect measures on

putative moderators and mediators of treatment response and report their findings, even when nonsignificant. This would allow for data pooling across studies and implementation of advanced techniques, such as meta-analytic structural equation modeling (e.g. [51]), in order to facilitate the identification of mechanisms of behavioral therapy.

4. Access to behavioral therapy treatment

Availability and access to in-person delivered behavioral therapy for tic disorders are beset by barriers which severely impede patient ability to obtain appropriate and effective treatment. Multiple studies [52–57] underline the difficulty of accessing behavioral therapy. Furthermore, one study [58] carried out in the UK found that BT is available to only 20% of the children and adolescents who require it. This study also reports that the quality of treatment is impaired by the fact that over 50% of the pediatric patients do not receive the recommended number of treatment sessions. Another study [59] found that notwithstanding the efficacy of CBIT, most children do not receive evidence-based treatment for tics.

4.1. Lack of therapists

One of the main barriers to the implementation of CBIT as the first-line treatment of tic disorders is the inadequate supply of appropriately trained and qualified CBIT therapists [52–57,59]. Efforts have been made to increase the provision of access to training and education for psychologists and occupational therapists, but in reality, many patients requiring CBIT treatment are unable to access it [60].

4.2. Geography and cost

When CBIT training is available, it is usually only at specialist treatment centers [56], which present problems for patients who have to travel long distances. Finding local providers is a common difficulty, and in such cases where geography necessitates long journeys for treatment, the time traveling, for some, constitutes a prohibitive cost burden [61]. In countries in which CBIT is not provided by the public health system, the cost of private behavioral therapy is beyond the means of many patients and their families [60].

4.3. Remote treatment increases availability

To overcome the hurdle of inadequate behavioral therapy Himle [53] proposed the need for ‘novel strategies.’ To that end, in the last decade efforts to make CBIT more available and cost-effective have centered on the use of video conferencing, internet-based self-help programs, and therapist supported online remote behavioral interventions. Video conferencing is a solution to accessibility issues related to travel or distance, but it does not efface the problem of a shortage of trained therapists as it is still reliant on their availability to conduct the video sessions [52]. However, a vital difference between internet-delivered behavioral therapy with minimal therapist involvement and face-to-face or video conference behavioral therapy is the reduction in the therapist

time required to deliver the treatment. The efficient use of therapist time should mean that for every person treated one on one, four people can be supported by online delivery, with the result of lower costs [56]. Overall, the delivery of treatment remotely is an approach which increases the availability of behavioral therapy [55].

4.3.1. Therapist-led remote interventions

In recent years, several studies have assessed the efficiency of remote behavioral therapies. There is one Class II study of CBIT using a voiceover Internet protocol (VoIP) compared to wait list controls in 20 children and youth [62]. Children received eight sessions of CBIT VoIP over the Internet over 10 weeks. Children receiving CBIT VoIP had a significant decrease in the YGTSS TTS at the end of the study, but there was no significant difference between the CBIT VoIP and the wait list control group at the endpoint (SMD, 0.24 [95% CI, –0.65, 1.14]). Another Class II study in 20 children with TS compared CBIT delivered face-to-face with CBIT delivered through telehealth [63]. Children received eight sessions of CBIT over 10 weeks either in person or by video conference. Both groups significantly improved in tic severity on the YGTSS TTS, with no difference between the methods of treatment administration at the endpoint (SMD, 0.24 [95% CI, –0.70, 1.17]). Finally, a group from Catania [64] conducted a Class IV RCT in which they compared behavioral therapy delivered through the videoconference to standard face-to-face treatment. Behavioral therapy consisted either of HRT or ERP conducted over eight weekly sessions. Forty children and adolescents were enrolled in this study. Both groups demonstrated not only improvement in tics, but also obsessive-compulsive symptoms and anxiety. The YGTSS score decreased by 11.4 (SD 6.6) points with online remote behavioral therapy, and 12.1 (SD 6.0) with face-to-face behavioral therapy, with no significant difference between groups. A study in the United States found that 25% of patient offered treatment chose to do CBIT over videoconference and rated the treatment as highly satisfactory [65]. Thus, evidence for the efficacy of therapist-led remote intervention is beginning to emerge and it suggests that it could be as effective as face-to-face, in-person intervention.

4.3.2. Studies of Internet delivered treatment with minimal therapist involvement

In order to improve accessibility to behavioral therapy, researchers have developed new treatment programs that include minimal therapist involvement.

Andren et al. [55] performed a single blind parallel group pilot Class II RCT in which they evaluated the feasibility of adapting two behavioral therapy protocols (HRT or ERP) delivered through a therapist-guided and parent-guided online self-help format. Twenty-three children with tic disorders were randomized to 10 weeks of therapist-guided and parent-guided internet-administered programmes of HRT or ERP. The primary outcome was the change in the YGTSS-TTS. The assessment of efficacy was done at baseline, after treatment and at 3-month follow-up. Importantly, all participants also completed a 12 month follow-up. Both interventions improved tic-related impairment, parent-rated tic severity, and quality of

life, but only ERP resulted in improvement of the YGTSS-TTS. Within-group effect sizes and responder rates were, respectively: $d = 1.12\%$ and 75% for ERP, and $d = 0.50\%$ and 55% for HRT. The effect of the therapy persisted over 12 months.

Following this pilot study, two identical Class II trials of therapist-supported internet delivered ERP were conducted in England and Sweden, with the study in England completed first. In England [56], 224 children were randomly allocated to either therapist-supported online remote ERP or online psychoeducation for 10 weeks. The primary outcome was the YGTSS-TTS assessed after 3 months. While both interventions led to tic improvement, ERP resulted in significantly better symptom amelioration (effect size of -0.31 (95% CI -0.52 to -0.10) compared to psychoeducation. Likewise in Sweden, [66] 221 children with tic disorders were randomized to either ERP or structured education over a period of 10 weeks. Tic severity was assessed according to the YGTSS-TTS and CGI-I scales. Both groups improved significantly in terms of tic severity with no between-group differences. However, the ERP group had a significantly higher treatment response rate (47% vs 29%).

Finally, a novel parent-guided internet-delivered CBIT (iCBIT) program was evaluated in a feasibility and effectiveness Class IV study [59]. Youths were randomly assigned to either iCBIT or a wait-list control group. The main objective of the iCBIT program was to provide the child with specific skills for managing tics, as well as to facilitate caregiver support. This intervention enabled the caregiver to present modules to the child, with only minimal weekly communication with assistants who provided feedback. iCBIT was associated with a mean YGTSS TTS reduction of 6.60 points ($p < 0.001$) compared with a reduction of 0.94 points ($p = 0.51$) in the wait-list group. All in all, emerging evidence suggests that internet-delivered treatment programs with minimal therapist involvement are effective in reducing tic severity.

4.3.3. Self-help programs

Online self-help programs and books are readily available for patients and families. TicHelper is an online tool based on the therapist-guided protocol for CBIT [67]. It is comprised of four modules in a stepped care model of treatment, which follows the key components of CBIT. Lasting 8 weeks, the content is addressed to parents to use with their children. TicHelper is easily accessible and, considering the length of treatment available, inexpensive. An expert review of TicHelper was positive, acknowledging it represents an accessible and low-cost alternative to face-to-face behavioral therapy [67], and noted that the content mirrors the CBIT therapist manual in a more easily understood form. Its core weakness, the reviewers underline, is that the narrow focus of the TicHelper on only reducing tics may have less of an effect on secondary psychiatric symptoms. Leaky Brakes constitutes another online self-help program option. It is a free, online video-based tic management resource, with a child-friendly adapted CBIT program. Eight videos comprise the Tic Management Toolbox. The video sessions range from 'creating a tic list' to 'using your tic blocker,'

and each session includes associated and practice materials, which are aimed at the adults needed to assist the child through the course. This resource is a practical and fun program, designed to engage youth, though the advised logging and recording of progress may prove difficult for a child without direct adult supervision.

Scientific evidence supporting the efficacy of such online self-help programs is, however, scarce. Haas et al. [68] recently published the results of their multicenter Class II RCT in which they evaluated the efficacy and safety of an internet-delivered, completely therapist-independent intervention (iCBIT Minddistrict®) for the treatment of tics in adults and compared it with online psychoeducation and classical face-to-face CBIT. The primary outcome was tic improvement on the YGTSS-TTS 1 week after treatment. In comparison to psychoeducation, iCBIT significantly decreased the YGTSS TSS 3 months [mean difference -2.25 (95% CI -3.75 , -0.75)] and 6 months [mean difference -2.71 (95% CI -4.27 , -1.16)] after treatment, but not 1 week after treatment [mean difference -1.28 (95% CI -2.58 , 0.01)]. Comparing iCBIT to face-to-face CBIT, there was no significant difference at any time point between groups, with both treatment groups demonstrating significant improvement over time.

4.3.4. Treatment manuals

Managing Tourette Syndrome: A Behavioural Intervention [8] is a manualized CBIT program developed by Douglas W. Woods specifically devised for therapists. This manual is particularly helpful and practical for therapists who have difficulty finding or attending behavioral therapy training courses. The guide contains step-by-step instructions for each session, including the materials to support the patient. The accompanying treatment workbook manuals for parents allows families to understand the principles of CBIT, follow their child's treatment, and assist with homework. Similarly, *Tics: Therapist Manual and Workbook for Children* by Verdellen et al. provides step-by-step instructions for therapists on ERP (12 sessions) and HRT (10 sessions) for children 7–13 years with tics. A workbook for parents is also available [69].

4.3.5. CBIT books for patients

Nix Your Tics! Eliminating Unwanted Tic Symptoms: A How-To Guide for Young People [70] is a book aimed at adolescents but is equally useful for adults. The book provides a short but engaging overview of the habit reversal therapy (HRT) and competing response components of CBIT. The book offers much support to both the individual using HRT as well as to supporting family members, emphasizing the importance of parental assistance in achieving the goal of tic reduction.

4.4. Training programs for therapists

The lack of therapists available and trained in CBIT is an important barrier to access to behavioral therapy and has been described in Europe [52], UK [57] and North America

[54]. Specialized CBIT training for therapists is required to support patients with behavioral therapy, and few training courses actually exist. Access to available training for many therapists is limited as training opportunities will almost certainly not exist in their community. An online search on CBIT training providers in North America results in fewer than a handful. In seeking to overcome this barrier, the Tourette Association of America organizes training programs in different city locations across America. Each program lasts 2 days, and the participants who complete the training have to carry out follow-up consultations with their trainer. Similarly, the Canadian Psychological Association offers a two-day virtual training program. The cost for training programs might present a barrier for therapists with funding limitations. Neurodiverse.org has a Behavioral Therapy for Tics Institute, which provides online training for clinical professionals which may be a more affordable option.

5. Expert opinion

There is good evidence to support the use of CBIT, face-to-face, and one-on-one with a therapist, with multiple Class I RCTs [10,12] finding it superior to supportive therapy, both in children and adults. Evidence from these trials led to clinical practice guideline recommendations for the use of CBIT as a first-line intervention in patients with TS and chronic tic disorder who are experiencing tic-related impairment, or who do not have impairment but are motivated to try this treatment [2]. These recommendations were made based on high confidence in the evidence for efficacy, and low risk of adverse effects from treatment. ERP also appears to be efficacious for tics but does not have the same foundation of evidence as CBIT, at least for face-to-face, one-on-one treatment compared to a control condition.

A few studies either used behavioral interventions which combine habit reversal therapy and ERP for tics [19] or used them interchangeably [21,64]. This is likely a greater approximation to real world behavioral treatment, where therapists tailor their treatment strategies depending on their individual patients' tics and treatment responses. The comparative study of behavioral (HRT or ERP), educational and pharmacological treatments is an important landmark, demonstrating that behavioral therapy is as effective as pharmacological therapy with antipsychotic medications in children and adolescents [21].

More recent studies have confirmed that a one-on-one treatment using video conferencing seems just as effective as face-to-face treatment [53,64]. This is reassuring and may allow greater convenience for families living in more remote communities, or for whom qualified practitioners are a distance away. Not surprisingly, group-based behavioral treatment does not appear as effective as one-on-one treatment, but only one trial has compared these two interventions [19]. Two trials comparing group-based behavioral treatment to group-based psychoeducation found a greater improvement in motor tics, but no improvement in phonic tics with behavioral treatment [15,17]. Of greater concern, in one of these trials, vocal tics increased in both groups

postintervention, suggesting a possible negative immediate influence of group treatment on vocal tic severity [17].

Internet-based behavioral treatment programs with minimal or no therapist involvement appear more effective than psychoeducation, although the magnitude of improvement appears smaller than traditionally administered treatment [66]. This again, is not unexpected, given the more self-directed and less intensive nature of internet-based treatment. People with tics showing inadequate treatment responses to internet-based behavioral treatments should therefore consider face-to-face or video-conferenced one-on-one treatment, if available, prior to concluding that behavioral treatment is ineffective.

Evidence on how and for whom behavioral therapies for tics work is just emerging, and it is difficult to form strong conclusions currently. At this time, the most important actionable finding is that treatment outcomes appear to be moderated by the number of sessions, with larger effects sizes in studies using more sessions [28,58]. As the majority of children with tics do not receive the recommended number of sessions, efforts should focus on encouraging adherence to treatment manuals in order to obtain the greatest benefit from treatment. The overall generalizability of findings from research studies may also be called into question, as the majority of behavioral therapy studies have occurred in North America, Western Europe, and Israel, with the majority of participants male, Caucasian, and coming from well-educated families.

Finally, while long-term data are scarce, the longest follow-up study of CBIT trial participants 11 years later suggests that CBIT improves the likelihood of tic remission [11]. This suggests that CBIT may alter the natural history of tic disorders, by teaching management strategies for tics which people can gain mastery of and use over years. This therapy should be considered a long term, worthwhile investment for individuals with tics and by payers.

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